

ST BRIEUC ARMOR AP, France

WMO: 071200

Lat: 48.5364N

Lon: 2.8532W

Elev: 453

StdP: 14.46

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.6	31.1	22.2	17.0	31.4	25.2	19.5	33.0	30.4	50.2	27.6	49.3	6.8	130	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.6	77.8	65.9	74.0	64.2	71.4	62.9	67.4	74.9	65.6	71.8	64.2	69.5	8.5	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
64.6	92.9	70.5	63.2	88.1	68.3	62.0	84.5	67.0	32.0	75.2	30.7	71.9	29.6	69.8	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.5	21.5	19.2	DB	24.4	86.7	4.3	5.0	21.2	90.3	18.7	93.2	16.3	96.0	13.1	99.6	(4)
(5)			WB	22.8	71.0	4.1	2.7	19.9	72.9	17.5	74.5	15.2	76.0	12.2	77.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	52.3	43.2	43.7	46.1	49.0	54.0	59.0	62.6	62.7	59.4	54.9	48.6	44.3	(6)
(7)		DBStd	8.63	5.95	5.39	5.06	5.00	4.68	4.65	3.85	3.95	4.36	4.90	4.77	6.02	(7)
(8)		HDD50	918	216	185	142	75	15	1	0	0	1	13	80	190	(8)
(9)		HDD65	4682	675	598	586	480	342	189	94	93	176	315	493	641	(9)
(10)		CDD50	1773	6	7	21	46	139	271	390	394	283	165	37	14	(10)
(11)		CDD65	61	0	0	0	0	1	9	19	22	9	1	0	0	(11)
(12)		CDH74	344	0	0	0	3	11	80	99	101	44	6	0	0	(12)
(13)		CDH80	70	0	0	0	0	1	20	19	23	6	1	0	0	(13)
(14)	Wind	WSAvg	9.5	11.1	11.0	10.3	9.7	9.0	8.5	8.5	8.1	8.6	8.9	9.9	10.5	(14)
(15)	Precipitation	PrecAvg	30.3	3.1	2.8	2.1	2.4	2.2	1.8	1.8	1.9	1.9	3.1	3.6	3.6	(15)
(16)		PrecMax	39.0	6.3	5.7	6.0	6.7	5.3	4.7	5.4	4.1	5.7	6.6	7.0	8.2	(16)
(17)		PrecMin	20.0	0.4	0.5	0.4	0.6	0.7	0.3	0.5	0.6	0.2	0.5	1.1	0.8	(17)
(18)		PrecStd	4.4	1.7	1.3	1.3	1.4	0.9	1.1	1.1	1.0	1.1	1.3	1.6	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	56.5	59.7	64.4	71.2	75.0	82.7	81.9	83.7	79.9	71.6	62.2	57.8	(19)
(20)			MCWB	53.8	52.3	53.9	57.7	62.0	68.5	67.8	68.4	65.1	62.7	58.8	55.1	(20)
(21)		2%	DB	54.9	55.2	59.0	64.3	68.4	74.9	76.2	76.4	73.2	66.6	59.5	56.3	(21)
(22)			MCWB	52.8	51.4	52.6	56.1	60.0	64.8	65.2	66.2	63.3	60.7	56.6	54.0	(22)
(23)		5%	DB	53.3	53.2	56.0	60.2	65.0	70.3	72.8	73.0	69.7	64.1	57.5	55.0	(23)
(24)			MCWB	51.1	50.4	51.7	53.5	58.1	61.9	63.7	64.6	62.0	59.5	54.9	53.0	(24)
(25)		10%	DB	51.6	51.6	53.8	56.8	62.0	67.5	70.2	70.0	66.8	62.1	55.7	53.3	(25)
(26)			MCWB	49.5	49.0	50.3	51.3	56.2	60.4	62.5	62.9	60.7	58.4	53.3	51.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.7	54.0	55.9	59.6	64.2	69.0	69.5	70.3	67.3	63.9	59.9	56.0	(27)
(28)			MCD B	56.1	57.2	60.7	66.9	71.7	79.3	79.6	79.3	73.8	68.5	61.4	57.3	(28)
(29)		2%	WB	53.2	52.4	54.0	56.3	61.2	65.6	66.6	67.5	65.0	62.0	57.1	54.5	(29)
(30)			MCD B	54.6	54.6	57.5	62.8	67.4	73.5	73.5	74.2	70.9	65.4	58.9	56.0	(30)
(31)		5%	WB	51.5	50.8	52.4	54.2	58.8	63.0	65.0	65.6	63.2	60.4	55.4	53.2	(31)
(32)			MCD B	53.0	52.9	55.5	59.2	63.5	68.8	71.0	71.0	68.0	63.3	57.4	54.9	(32)
(33)		10%	WB	49.9	49.2	50.8	52.2	57.0	61.1	63.6	64.1	61.7	59.0	53.6	51.4	(33)
(34)			MCD B	51.4	51.3	53.7	56.1	61.2	66.2	69.0	68.5	65.8	61.5	55.5	53.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	8.7	9.9	11.5	13.1	13.6	14.5	14.5	14.6	13.9	11.4	9.4	8.8	(35)
(36)			MCDBR	9.4	11.5	14.2	17.6	17.6	19.3	18.5	18.7	17.5	13.1	10.5	9.2	(36)
(37)			MCWBR	8.7	8.9	9.5	10.6	10.6	10.8	9.6	9.7	9.2	8.3	8.7	8.5	(37)
(38)		5% WB	MCD BR	9.2	10.8	13.0	16.1	16.5	17.9	16.8	16.6	16.0	12.0	10.2	9.3	(38)
(39)			MCWBR	8.9	9.2	9.3	10.5	10.7	10.8	9.7	9.6	9.5	8.6	9.0	9.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.326	0.342	0.371	0.384	0.389	0.384	0.382	0.372	0.359	0.353	0.331	0.320	(40)	
(41)		taud	2.417	2.389	2.314	2.294	2.323	2.364	2.373	2.422	2.440	2.452	2.441	2.390	(41)	
(42)		Ebn at Noon	233	256	266	274	277	278	277	275	268	249	229	217	(42)	
(43)		Edh at Noon	21	27	34	38	39	37	36	33	30	25	21	19	(43)	
(44)	All-Sky Solar Radiation	RadAvg	305	552	890	1369	1624	1786	1703	1446	1146	666	372	257	(44)	
(45)		RadStd	27	63	91	134	133	198	104	117	84	55	38	32	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page